

QuillNet: Query Reformulation Without Supervised Pairs

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Abstract

We present QuillNet, a self-supervised reformulation approach to open-domain question answering. Existing systems treat this problem as a single monolithic decision, which wastes computation on inputs that are easy and starves the inputs that are hard. QuillNet instead allocates effort where the evidence is ambiguous. On AskCorpus we reach 78.2 recall@20, against 71.0 recall@20 for the TermExpand baseline, while remaining trainable on four commodity CPU workers.

1 Introduction

Work on open-domain question answering has converged on a small number of architectures that differ more in their training recipes than in their inductive biases. Progress reported on public leaderboards is therefore difficult to attribute, and practitioners choosing a system for a modest deployment have little guidance beyond parameter count. We take the opposite starting point and ask what can be achieved when the compute budget is fixed in advance at four commodity CPU workers.

Our contribution is QuillNet, which reorganises open-domain question answering around the observation that most inputs are decided by a small fraction of the available evidence. Rather than scaling the model, we make the decision procedure adaptive. This is a deliberately unfashionable direction: it trades headline capacity for predictable cost, and it is the trade practitioners actually face.

We evaluate on AskCorpus, a collection of 112,300 examples assembled for open-domain question answering. Against the TermExpand baseline, QuillNet improves recall@20 from 71.0 to 78.2. An ablation shows that disabling the round-trip consistency filter costs 6.1 points of recall@20, which locates the gain in the component we claim is responsible rather than in incidental tuning.

The remainder of this paper is organised as follows. Section 2 describes QuillNet in detail. Section 3 sets out the experimental protocol and the baselines we compare against. Section 4 reports results and ablations. Section 5 discusses the limitations of the approach, of which the most serious is that the reformulator inherits the lexical biases of the corpus it was bootstrapped from, and transfers poorly to a new domain without a fresh bootstrap.

2 Method

QuillNet has three stages: a cheap scorer that produces a provisional decision, a confidence estimator that decides whether the provisional decision is safe, and an expensive scorer that is invoked only when it is not. The three stages share an encoder, so the additional parameter cost over the TermExpand baseline is under four percent.

The confidence estimator is the component that makes the self-supervised reformulation framing pay. We train it with a margin objective against the disagreement between the cheap and expensive scorers, which requires no additional human annotation: the supervision is manufactured from the two scorers that already exist.

Training proceeds in two phases. The cheap and expensive scorers are trained jointly for the first phase, then frozen while the estimator is fitted. Freezing matters. When all three are trained end to end the estimator learns to mark everything as uncertain, because that minimises loss without ever being penalised for the compute it spends.

Inference is a single forward pass for the majority of inputs. On AskCorpus the expensive scorer fires on roughly one input in five, which is what makes QuillNet runnable on four commodity CPU workers at all. We provide the exact thresholds and the schedule used for every reported number in the supplementary material.

3 Experiments

We evaluate on AskCorpus, which contains 112,300 examples. We use the standard split released with the dataset and report recall@20 on the held-out test portion. No test example is used for threshold selection; thresholds are chosen on a validation split carved out of the training portion.

Our primary baseline is TermExpand, which we retrain under our own protocol rather than quoting published numbers. Retraining moves the TermExpand figure by more than a point relative to the number reported by its authors, which is itself a small argument for retraining baselines rather than copying tables.

All runs use four commodity CPU workers. Every reported figure is the mean of five seeds. We report the mean rather than the best seed because the seed-to-seed spread on AskCorpus is roughly one point of recall@20, which is large enough that a best-of-five protocol would manufacture most of a claimed improvement.

We additionally run an ablation in which disabling the round-trip consistency filter is applied, holding every other setting fixed. This isolates the contribution of the component we claim is doing the work, and it is reported alongside the main result in the next section.

4 Results

QuillNet reaches 78.2 recall@20 on AskCorpus. The TermExpand baseline reaches 71.0 recall@20 under the identical protocol. The gap is stable across all five seeds and does not depend on the choice of validation split.

The ablation confirms where the gain comes from. With disabling the round-trip consistency filter, recall@20 falls by 6.1 points, recovering most of the distance back to TermExpand. We read this as evidence that the improvement is a property of the self-supervised reformulation design rather than of the training schedule.

Cost is the other half of the claim. QuillNet completes the full AskCorpus test pass on four commodity CPU workers, which the uncascaded configuration does not. For a team without cluster access this is the difference between a system that can be evaluated and one that can only be cited.

5 Limitations

The most serious limitation of QuillNet is that the reformulator inherits the lexical biases of the corpus it was bootstrapped from, and transfers poorly to a new domain without a fresh bootstrap. We report this prominently because it determines whether the method is appropriate for a given deployment, and it is not visible from the headline recall@20 figure.

A second limitation is evaluation scope. AskCorpus is a single dataset in a single language, and we make no claim that the 6.1 point ablation gap would reproduce elsewhere. Cross-dataset transfer is the obvious next study and we have not run it.

References

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